

Sp 23 ANS

1a

The Linear DC Machine as a Motor:

Suppose that, the linear machine is initially running at the no-load steady state condition.

When an external load is applied, then it will be figure.

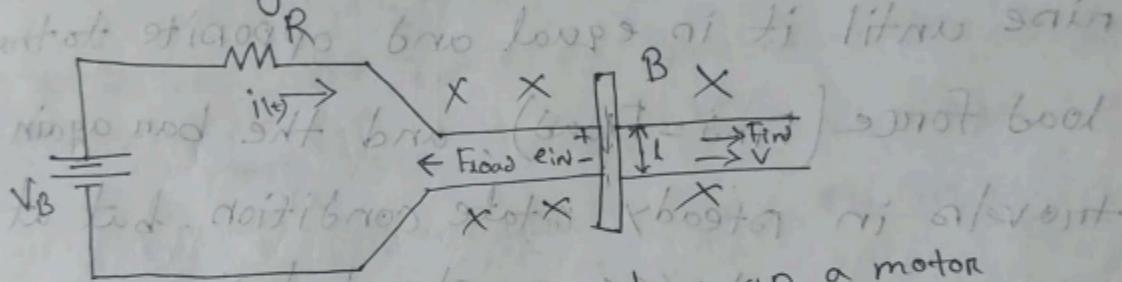


Fig: Linear dc machine as a motor to the bar

A force F_{load} is applied opposite to the direction of motion, which causes a net force F_{net} on the bar in the direction opposite to the direction of motion.

$$F_{net} = F_{load} - F_{ind}$$

So, the bar will slow down ($a = F_{net}/m$) in magnitude. As a result, the induced voltage on the bar

drops ($\epsilon_{ind} = vBl$)

☐ When the induced voltage drops,
the current flow in the bar will rise

$$i \uparrow = \frac{v_B - \epsilon_{ind} \downarrow}{R} \quad | \quad v_B = iR + \epsilon_{ind}$$

☐ Thus, the induced force will rise too.

$$(F_{ind} \uparrow = i \uparrow lB)$$

1b

Hysteresis Loss	Eddy Loss
1) The losses which occur because of the <u>reversal of the magnetism</u> is known as the hysteresis loss.	1) The loss occurs because of the <u>relative motion</u> between the core and the magnetic flux is known as the Eddy Loss.
2) Formula: $P_h = K_h f B_{max}^n V$ $P_h = K_h f^2 B_{max}^2 V$	2) Formula: $P_e = K_e f^2 B_{max}^2 n^2 V$ $P_e = K_e f^2 B_{max}^2 V$
3) <u>occurs</u> The hysteresis loss occurs because of the reversal of the magnetism. 4) Minimized by the Silicon steel.	3) <u>occurs</u> interaction of magnetic field and conductor. 5) Minimized by using the thin core of Lamination.

Motor sizing refers to the process of picking the correct motor for a given load. It is important to size a motor correctly because:

- If a motor is **too small** for an application it may not have sufficient torque to start the load and run it up to the correct speed. Even if it does get the load up to speed the motor will **overheat** and **burnout** if it is too small for the application.
- If a motor is **too large** for an application then money has been wasted in purchasing such a large motor. Also motors typically operate inefficiently when they are run well below rated load. **Some** money is also wasted in running costs.

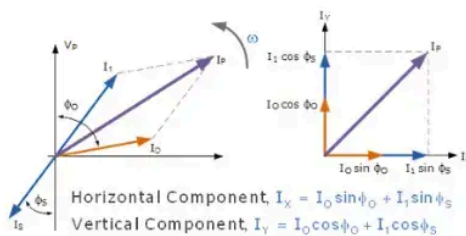
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- For the load torque is constant then choosing a motor whose rated torque is slightly above the torque required by the load. The **load torque should be between 75%-100%** of the rated motor torque with **95%** being an ideal choice.
- For constant speed applications where the motor will run continuously at rated speed you do not need to work out torque. You can simply look at **load power** and **motor power** and use the same 75%-100% rule. This is the case for standard induction motors without a variable speed drive.

2a

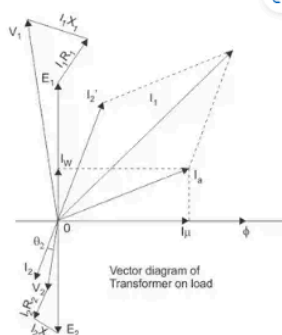
What is the Vector Diagram of Transformer?



$$I_p = \sqrt{I_X^2 + I_Y^2} \text{ and } p.f. = \cos \phi = \frac{I_Y}{I_p}$$



Electrical 4 U



Working principle of transformer

The transformer works on the principle of [Faraday's law](#) of electromagnetic induction and mutual induction.

There are usually two coils – primary coil and secondary coil – on the transformer core. The core laminations are joined in the form of strips. The two coils have high mutual inductance. When an alternating current passes through the primary coil, it creates a varying magnetic flux. As per Faraday's law of electromagnetic induction, this change in magnetic flux induces an EMF (electromotive force) in the secondary coil, which is linked to the core having a primary coil. This is mutual induction.

Overall, a transformer carries out the following operations:

1. Transfer of electrical energy from one circuit to another
2. Transfer of electrical power through electromagnetic induction
3. Electric power transfer without any change in frequency
4. Two circuits are linked with mutual induction

2b



2B

A 50-KVA 4400/220 V transformer has $R_1 = 3.45 \Omega$, $R_2 = 0.009 \Omega$. The values of reactances are $X_1 = 5.2 \Omega$ and $X_2 = 0.015 \Omega$. Calculate for the transformer.

(i) equivalent resistance as referred to primary.

Ans

$$R_{01} = R_1 + R_2'$$

$$= R_1 + \frac{R_2}{K^2}$$

$$= 3.45 + \frac{0.009}{(0.05)^2}$$

$$= 7.05 \Omega$$

$$K = \frac{220}{4400} = 0.05$$

$$R_1 = 3.45$$

$$R_2 = 0.009$$

$$X_1 = 5.2$$

$$X_2 = 0.015$$

(ii) Equivalent resistance as referred to secondary.

$$R_{02} = R_1' + R_2 = 220.0 + (3.45 \times 2) =$$

$$= R_1 \times K^2 + R_2$$

$$= 3.45 \times (0.05)^2 + 0.009$$

$$= 0.176 \Omega$$

(iii) equivalent ~~resistance~~ both primary and Secondary reactance

$$X_{01} = X_1 + \frac{X_2}{K^2} = 5.2 + \frac{0.035}{(0.05)^2} = 11.2 \Omega$$

$$X_{02} = X_2 + X_1 \cdot K^2 = 0.035 + 5.2 \times (0.05)^2 = 0.028 \Omega$$

(iv) Equivalent impedance as referred to both primary and Secondary

Find value in (3) (a)

$$\text{Impedance as Primary } Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{(7.05)^2 + (11.2)^2}$$

$$= 13.23 \Omega$$

$$\text{Impedance as Secondary } Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

$$= \sqrt{(0.176)^2 + (0.028)^2}$$

$$= 0.176 \Omega$$

2c

Let's go through each of the requested calculations based on the given information for the no-load test of the transformer:

Given:

- Primary voltage (V_p) = 220V
- Secondary voltage (V_s) = 110V
- Power input (P) = 30W
- Primary current (I_p) = 0.5A

****1. Turn Ratio (N_p/N_s):****

The turn ratio of a transformer is the ratio of the number of turns in the primary winding (N_p) to the number of turns in the secondary winding (N_s). Since the number of turns is not provided, we can calculate the turn ratio based on the voltage ratio:

$$\text{Turn Ratio } (N_p/N_s) = V_p / V_s = 220V / 110V = 2$$

****2. Magnetizing Component of No-Load Current (I_m):****

The magnetizing component of the no-load current is the current that flows through the primary winding to establish the magnetic flux in the core. Given the primary current (I_p) and power input (P), we can calculate the magnetizing current:

$$\text{Apparent Power } (S) = V_p * I_p = 220V * 0.5A = 110VA$$

$$\text{Magnetizing Current } (I_m) = S / V_p = 110VA / 220V = 0.5A$$

****3. Working Component of No-Load Current (I_w):****

The working component of the no-load current accounts for the losses in the core and is the difference between the total no-load current and the magnetizing current:

Total No-Load Current (I_o) = I_p

Working Component of No-Load Current (I_w) = $I_o - I_m = 0.5A - 0.5A = 0A$

****4. Iron Loss (Core Loss):****

The iron loss, also known as core loss, is the power consumed by the transformer due to hysteresis and eddy current losses in the core material. Since the power input (P) is given as 30W, and the iron loss is the difference between the power input and the power used to establish the magnetizing current:

Iron Loss = Power Input - Power Used by Magnetizing Current

Iron Loss = $30W - (V_p * I_m) = 30W - (220V * 0.5A) = 30W - 110W = -80W$

3a

basic prenciple of single loop dc generator

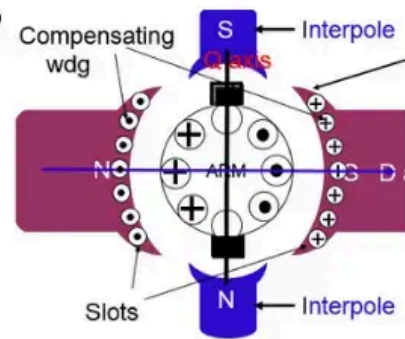
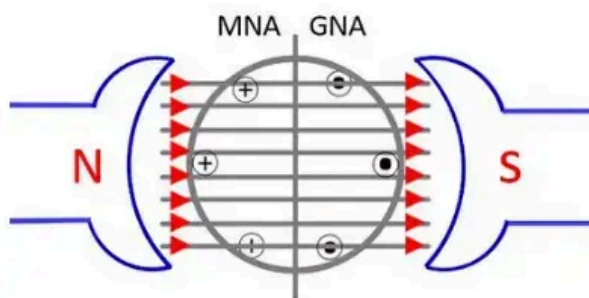
The current induced in the coil is collected and conveyed to the external load circuit by connecting the coil terminals to two continuous and insulated rings, known as slip rings or collector rings, mounted on the generator shaft and making the two stationary brushes pressing against the slip rings, one coil is rotated, the generated **alternating emf** causes a current to flow first in one direction and then in the other through the coil and external circuit.

3b

In a DC machine, the carbon brushes are always placed at the magnetic neutral axis. In no load condition, the magnetic neutral axis coincides with the geometrical neutral axis. Now, when the machine is loaded, the armature flux

is directed along the inter polar axis (the axis in between the magnetic poles) and is triangular in wave shape. This results an armature current flux directed along the brush axis and causes cross magnetization of the main field. This cross magnetization effect results in the concentration of flux at the trailing pole tip in generator action and at the leading pole tip in motor action. The armature reaction is the effect of the armature flux on the main flux. In case of a **DC motor** the resultant flux is strengthened at the leading pole and weakened at the trailing pole tips.

What is Armature Reaction in DC Machine?



Electrical 4 U